Lab 4 — Production ML Monitoring

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Objective
Instrument a credit-risk FastAPI service for operational, data-drift, fairness, and explanation signals. The stack exposes Prometheus metrics, seven alert rules, a provisioned Grafana dashboard, and SHAP explanations.

Architecture
Client → FastAPI middleware → scoring model → bounded monitoring window. Prometheus scrapes /metrics every 10 seconds; Grafana reads Prometheus; alert rules cover availability, errors, latency, drift, fairness, review volume, and stale model age.

Method
Each profile below sent 400 requests through the real FastAPI routes using TestClient. The monitoring window was reset between profiles so the comparison is not contaminated by previous traffic. PSI thresholds are stable < 0.10, moderate 0.10–0.25, and significant ≥ 0.25.

Source: reports/profile_results.json (generated, reproducible evidence)

Measured traffic profiles

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Profile	p95 ms	Mean score	PSI max	Fairness gap	Drift
normal	26.1	0.236	0.030	0.019	stable
drifted	25.8	0.590	4.251	0.038	significant
unfair	26.3	0.397	0.353	0.722	significant

Interpretation

Normal traffic provides the operational baseline. The drifted profile deliberately moves age, credit limits, utilisation, repayment amounts, and delay status; PSI should therefore cross the significant threshold. The unfair profile changes only group 1's repayment behaviour, widening the selection-rate gap. These are investigation signals, not automatic proof that the model is incorrect.

Normal



Drifted



Unfair



Operational response and limitations

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Response playbook

- Availability/error alerts: inspect API logs and recent deploys; roll back if correlated.
- High latency: inspect request mix, CPU/memory, and explanation traffic separately.
- Significant PSI: identify the moving features, confirm upstream data contracts, and evaluate a labelled recent cohort before retraining.
- Fairness gap: review sample size and group composition, run outcome-based fairness analysis when labels mature, and involve model-risk/compliance owners.

Limitations

PSI is univariate and cannot detect relationship shifts. Selection-rate parity is a proxy and is not equal to outcome fairness. In-memory windows reset when the service restarts and are per replica. Production should persist observations or aggregate them centrally. SHAP explains model behaviour, not causality.

Verification

The automated suite contains 79 tests and reached 91.3% application-code coverage. GitHub Actions validates Prometheus rules, Docker Compose, the API suite, the full monitoring stack, and an end-to-end Prometheus scrape.